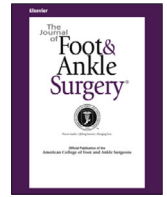




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The Relationship Between Fibular Notch Anatomy and ATFL Rupture

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ABSTRACT

The anterior talofibular ligament (ATFL) is the first to be damaged during a lateral ankle sprain. Dynamic and static structures have been investigated to better understand ATFL rupture, but the predisposing factors have not been fully elucidated. This study aims to define the fibular notch version that can evaluate the position of the fibular notch relative to the tibia and investigate the relationship between the fibular notch version (FNV) and ATFL rupture. This study included 71 patients with isolated ATFL rupture diagnosed clinically and radiologically and 71 control patients without any foot or ankle pathologies. Anterior facet length (AFL), posterior facet length (PFL), anterior-posterior facet angle (APFA), fibular notch depth (ND), and FNV measurements were performed on axial magnetic resonance images (MRI). We defined FNV as a parameter that evaluates the fibular notch's relative position to the distal tibia. The mean FNV was $16.6^\circ \pm 4.9^\circ$ in patients with ATFL rupture and $12.4^\circ \pm 5.6^\circ$ in the control group; when both groups were compared, FNV measurements were significantly higher in patients with ATFL rupture ($p = .002$). The mean APFA was $123.9^\circ \pm 10^\circ$ in the group with ATFL rupture and $129.7^\circ \pm 7.8^\circ$ in the control group. When both groups were compared, APFA was significantly lower in patients with ATFL rupture ($p = .014$). There was no significant difference between the groups regarding AFL, PFL, and ND. A more posterior (retroverted) fibular notch and a lower fibular notch angle seem to be associated with higher rates of ATFL rupture.

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Ankle sprains are among the most common sports injuries (1–3). The mechanism of injury, which is a combination of inversion and adduction in plantar flexion of the foot, can cause damage to the lateral ankle ligaments (1,4). The anterior talofibular ligament (ATFL) is the first ligament damaged during a lateral ankle sprain (5,6). In addition, it has been shown that after the ATFL is ruptured, the amount of internal rotation increases significantly, and the remaining intact ligaments are further strained (6).

Studies have shown that anatomical variations are associated with ankle instability. It has been demonstrated in previous studies that tibial plafond varus deformity is associated with ankle instability (7–9). Frigg et al (10) found chronic ankle instability associated with a larger talus radius and a smaller tibiotalar contact area. Scranton et al (11) showed that the fibula is located more posteriorly in patients with ankle instability, and they defined the axial malleolar index to evaluate the variations in the position of the fibula relative to the medial malleolus. In later studies, similar results were obtained with the study of

Scranton et al (12–14). Based on these studies evaluating ankle instability and the fibula's relative position to the tibia and talus, the fibular notch's relative position to the tibia was thought to be associated with ankle instability. This study hypothesizes a relationship between the fibular notch's relative position to the tibia and ATFL rupture.

This study aimed to describe the fibular notch version and evaluate the relationship between ATFL rupture and fibular notch version, anterior facet length, posterior facet length, anterior-posterior facet angle, and notch depth.

Patients and Methods

Seventy-one patients with isolated ATFL rupture on their MRIs who applied to the Necmettin Erbakan University Meram Medical Faculty Orthopedics and Traumatology Department outpatient clinics between 2014 and 2018 due to lateral ankle instability and met the inclusion criteria of the study, were included in the study. The patients included in the study were those with ankle sprains due to inversion and hyperflexion. Clinical examinations of the patients were performed after 48 hours of resting cold application and elevation, as defined by ISAKOS (International Society of Arthroscopy, Knee Surgery, and Orthopedic Sports Medicine) (15,16). In the clinical examination, the lateral ligament injury classification used by Malliaropoulos et al (17) was utilized, and the patients were determined as grade II [5–10 degrees reduced ROM (range of motion), 0.5–2 cm edema, normal stress radiographs]. In addition, MRI scans of 71 patients without ankle instability, who underwent ankle magnetic resonance imaging (MRI) for other reasons and were evaluated as normal, were retrospectively analyzed. Patients with additional ligament

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damage, lower extremity surgery, lower extremity deformity [(patients with leg length inequality, genu varum, genu valgum, genu recurvatum, hip and knee range of motion limitation in physical examination), foot deformities (hallux valgus (hallux valgus angle $>15^\circ$, intermetatarsal angle $>9^\circ$), pes cavovarus (talocalcaneal angle $<20^\circ$, calcaneal inclination $>30^\circ$, lateral talo-first metatarsal angle $>4^\circ$), pes planus (calcaneal inclination $<20^\circ$, lateral talocalcaneal angle $>45^\circ$, lateral talo-first metatarsal angle $>4^\circ$)] neuromuscular disease, hematological disease, congenital disease, as well as patients under the age of 18 and over 60 were excluded.

Ebraheim et al (18) defined the measurements of the fibular notch 1 cm above the level of the tibiotalar joint in axial computed tomography (CT) sections. There are also studies evaluating pathologies of the distal tibiofibular joint using magnetic resonance imaging (MRI) (19,20). Yıldırım et al (21) on the other hand, used fibular notch measurements defined by Ebraheim et al 1 cm proximal to the tibiotalar joint level in axial MRI sections. In our study, fibular notch measurements were also made 1 cm proximal to the tibiotalar joint in the axial MRI. The anterior facet length connecting the deepest point of the fibular notch and the Tillaux-Chaput tubercle type (Fig. 1), the posterior facet length connecting the deepest point of the fibular notch and the apex of the posterior Volkmann tubercle (Fig. 1), the angle between the anterior-posterior facet (Fig. 1), and the fibular notch depth (Fig. 2) were measured.

We defined the fibular notch version (FNV) in reference to the sagittal axis of the tibia with a single measurement. The FNV can evaluate the relationship between the Tillaux-Chaput tubercle forming the anterior of the fibular notch and the Volkmann tubercle forming the posterior of the notch.

The first step in FNV measurement is to draw the largest possible circle between the anterior and posterior cortices in the axial MRI section of the distal tibia and to determine the largest distance between the anterior cortex and the posterior cortex. The second step is to draw tangent lines at the points where this circle contacts the cortices anteriorly and posteriorly (these tangent lines are lines perpendicular to the circle diameter line passing through the center of the circle and giving the longest anterior-posterior distance). The third step is to determine the distal tibia sagittal axis by combining the circle's anterior and posterior tangent points. The last step is the angle (α) FNV between the line showing the sagittal axis of the tibia and the line connecting the apex of the Tillaux-Chaput and Volkmann tubercle (Fig. 3).

FNV, anterior facet length (AFL), posterior facet length (PFL), anterior-posterior facet angle (APFA), and notch depth measurements (ND) on 142 patients with standard ankle MR images were performed by 2 different observers in 2 separate sessions blinded to each other. Mean values of the 4 measurements with confidence intervals (95% CI) were presented.

The sample size was calculated using G*Power version 3.1.9.4 by entering mean and standard deviations. It was necessary to have at least 64 patients in each group in the calculation by accepting the effect size as 0.5, alpha 5%, and power at least 80%. The power of the study was rechecked and confirmed after statistical analysis (post-hoc).

Data analysis was performed using the SPSS software (IBM-SPSS 22.0, Armonk, NY). Descriptive statistics and frequency analysis were applied. Intra-observer and inter-observer reliability was tested by calculating Cronbach's alpha for each measurement

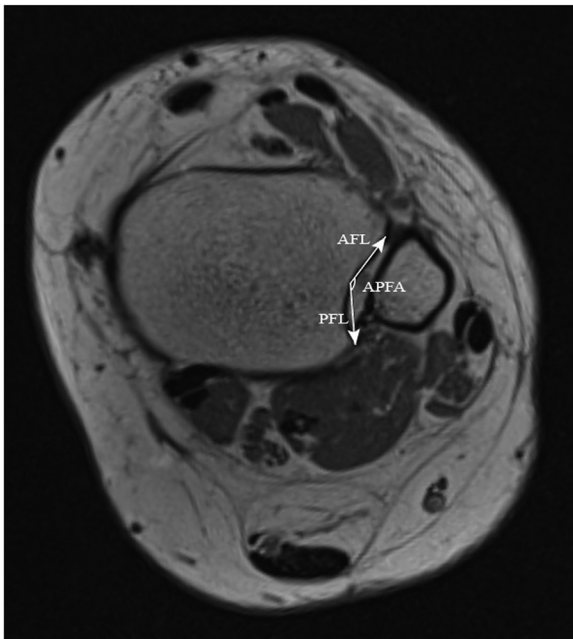


Fig. 1. Anterior facet length (AFL): measurement of anterior facet length, posterior facet length (PFL): measurement of posterior facet, anterior-posterior facet angle (APFA): measurement of anterior and posterior facet angle.

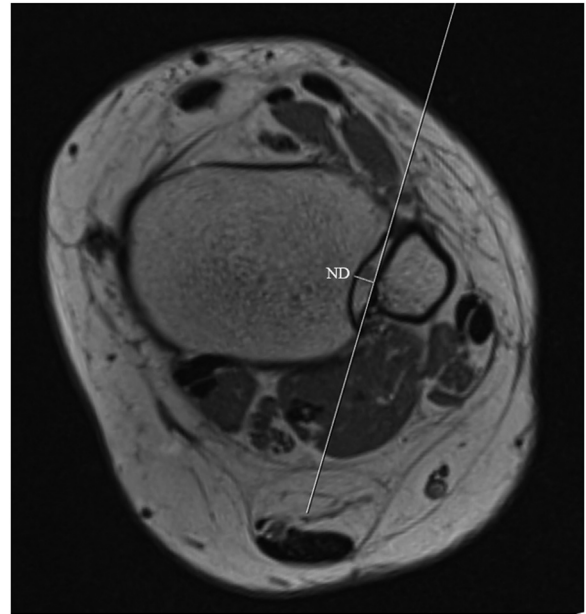


Fig. 2. Determination of notch depth (ND) is the distance between the line connecting the apex of the Tillaux-Chaput and Volkmann tubercle (L_4) and the deepest point of the notch.

repeated by a single investigator and between two investigators. The differences between the measured values and the means were calculated. The normality of the data was analyzed with the Shapiro-Wilk test. Mann-Whitney U and t-test were used to compare independent variables. Values with a $p < .05$ were considered significant.

Results

In the group with intact ATFL, the intrarater Cronbach's alpha values for AFL, PFL, APFA, and ND were 0.93, 0.91, 0.88, and 0.94, respectively, and the inter-rater Cronbach's alpha values were 0.91, 0.89, 0.85, and 0.89, respectively. While in the group with ATFL injury, the intra-rater Cronbach's alpha values for AFL, PFL, APFA, and ND were 0.91, 0.87, 0.93, and 0.90, respectively, and the inter-rater Cronbach's alpha values were 0.84, 0.89, 0.91, and 0.89, respectively (Table 1).

In the ATFL ruptured group, Cronbach's alpha analysis for FNV was intrarater 0.92, inter-rater 0.85, and the mean difference between observer-1 and observer-2 measurements was 1.44 ± 0.65 mm (95%CI: 1.18-1.69). While in the intact ATFL group, Cronbach's alpha analysis for FNV was intrarater 0.90, inter-rater 0.84, and the mean difference between observer-1 and observer-2 measurements was 2.94 ± 0.69 mm (95%CI: 1.71-2.17) (Table 1).

The mean age of 71 patients with ATFL rupture was 36.55 ± 11.6 y (min-max: 21- 57). The mean age of 71 patients in the control group was 33.7 ± 10.45 y (min-max:18-58). Demographic characteristics are summarized in Table 2.

When both groups of patients were compared regarding APFA, the APFA value of patients with ATFL rupture was significantly lower than those with intact ($p = .014$, power = 0.97, effect size = 0.64) (Table 3). The mean APFA value of patients with ruptured ATFL was $123.9^\circ \pm 10^\circ$ (range 100.3° - 138°) (95% CI: 121.58-126.22), while the mean APFA value of patients with intact ATFL was $129.7^\circ \pm 7.8^\circ$ (range 112° - 141.4°) (95% CI: 127.89-131.51) (Tables 3 and 4).

When both groups of patients were compared in terms of FNV, the FNV value of patients with ruptured ATFL was significantly higher than those who were healthy ($p = .002$ power: 0.99, effect size: 0.79) (Table 3). The mean FNV value of patients with ruptured ATFL was $16.6^\circ \pm 4.85^\circ$ (10.7° - 25.5°) (95% CI: 14.79-18.42), while the mean FNV value

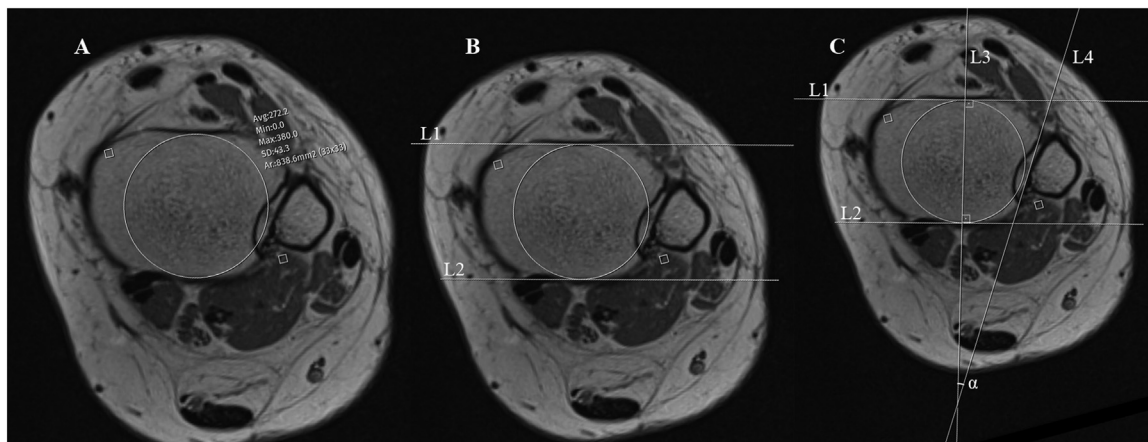


Fig. 3. The first step in fibular notch version (FNV) measurement (A) is to draw the largest possible circle between the anterior and posterior cortices in the axial MRI section of the distal tibia. The second step (B) is to draw tangent lines (L1 and L2) at the points where this circle contacts the cortices anteriorly and posteriorly (These tangent lines are perpendicular to the circle diameter line passing through the center of the circle and giving the longest anterior-posterior distance). The third step (C) is determining the distal tibia sagittal axis (L3) by combining the anterior and posterior tangent points of the circle and the angle between the line showing the sagittal axis of the tibia and the line connecting the apex of Tillaux-Chaput and Volkman tubercle (L4) is the FNV angle (α).

of patients with intact ATFL was $12.4^\circ \pm 5.04^\circ$ (0.7° – 19.0°) (95% CI: 10.88–14.40). (Table 3 and 4)

There was no significant difference between the two groups in terms of AFL ($p = .31$, power <0.80 , effect size $d:0.29$, for ruptured ATFL 95% CI:8.9–9.7, for intact ATFL 95% CI: 8.88–9.92), PFL ($p = .289$, power <0.80 , effect size $d: 0.27$, for ruptured ATFL 95% CI:13.08–14.12, for intact ATFL 95% CI: 13.67–14.93), and ND ($p = .105$, power <0.80 , for ruptured ATFL 95% CI: 4.37–4.87, for intact ATFL 95% CI: 3.99–4.55) (Table 3 and 4).

Discussion

The most important finding of this study is that ATFL rupture is associated with lower APFA values (deeper fibular notch) and higher FNV values (more retroverted fibular notch).

In our study, APFA was significantly lower in patients with ruptured ATFL with an associated more concave fibular notch. Accordingly, a distal fibula located in a deeper notch may be a risk factor for ATFL rupture. To the best of our knowledge, our study is the first in the literature to examine the relationship between APFA and ATFL rupture. The possible reason for this may be the limitation of anterior translational movement of the distal fibula in ankle plantar flexion and internal rotation of the distal fibula. Therefore, a more concave notch possibly results in premature stretching of the ATFL due to the limitation of anterior translational movement. When viewed from another angle, the high fibular notch anterior-posterior facet angle and low fibular notch depth (shallower fibular notch) may provide the opportunity for more internal rotation of the ankle, with plantar flexion and internal rotation of the ankle, and greater anterior translation of the distal fibula by the ATFL. Accordingly, the tensile force required for the rupture of the ATFL

may be delayed. Therefore, we think that a fibular notch with a low anterior-posterior facet angle and high depth increases the risk of ATFL rupture with its restrictive effect on the anterior translation of the distal fibula in plantar flexion and internal rotation. Furthermore, higher values of FNV increase the risk of ATFL rupture due to its restrictive effect on the internal rotation of the distal fibula in ankle plantar flexion and internal rotation. In our study, the fibular notch depth was higher in patients with ATFL rupture, but no significant relationship was seen between fibular notch depth and ATFL rupture.

Table 2

Demographic characteristics of patients (ATFL, Anterior talofibular ligament; M, Male; F, Female)

	ATFL Ruptured	ATFL Intact	<i>p</i>
Number of patients (n)	71	71	
Mean age $\pm$ SD (min-max)	36.55 $\pm$ 11.6 (21–57)	33.7 $\pm$ 10.45 (18–58)	.91
Sex (M/F)	31/40	34/37	

Table 3

Comparison of fibular notch version (FNV), anterior-posterior facet angle (APFA), anterior facet length (AFL), posterior facet length (PFL), and notch depth (ND) between patients with anterior talofibular ligament (ATFL) rupture and control group. (SD, standard deviation; CI, confidence interval; mm, millimeter; *, $p < 0.05$)

Radiological Parameter	ATFL		<i>p</i>
	Ruptured	Intact	
FNV (mean $\pm$ SD) (min-max) (95% CI)	16.6 $\pm$ 4.85 $^\circ$ (10.7 $^\circ$ –25.5 $^\circ$) (14.79–18.42)	12.64 $\pm$ 5.04 $^\circ$ (0.7 $^\circ$ –19 $^\circ$) (10.88–14.4)	.002*
APFA (mean $\pm$ SD) (min-max) (95% CI)	123.9 $\pm$ 10 $^\circ$ (100.3 $^\circ$ –138 $^\circ$) (121.58–126.22)	129.7 $\pm$ 7.8 $^\circ$ (112 $^\circ$ –141.4 $^\circ$) (127.89–131.51)	.014*
AFL (mean $\pm$ SD) (min-max) (95% CI)	9.3 $\pm$ 1.74mm (6.9–12.1) (8.9–9.7)	9.4 $\pm$ 2.23mm (6.7–14.5) (8.88–9.92)	.310
PFL (mean $\pm$ SD) (min-max) (95% CI)	13.6 $\pm$ 2.25mm (10.8–16.5) (13.08–14.12)	14.3 $\pm$ 2.7mm (9.6–18.9) (13.67–14.93)	.289
ND (mean $\pm$ SD) (min-max) (95% CI)	4.62 $\pm$ 1.07mm (3.24–7.14) (4.37–4.87)	4.27 $\pm$ 1.22mm (2.09–7.05) (3.99–4.55)	.105

Table 1

Reliability analyses of studied parameters

ATFL		
Radiological Parameter	Ruptured	Intact
FNV (intrarater / inter-rater cronbach alpha)	0.92/0.85	0.90/0.84
AFL (intrarater / inter-rater cronbach alpha)	0.91/0.84	0.93/0.91
PFL (intrarater / inter-rater cronbach alpha)	0.87/0.89	0.91/0.89
APFA (intra-rater / inter-rater cronbach alpha)	0.93/0.91	0.88/0.85
ND (intrarater/inter-rater cronbach alpha)	0.90/0.89	0.94/0.89

Table 4

Mean differences, CI values, and effect sizes for fibular notch version (FNV), anterior-posterior facet angle (APFA), anterior facet length (AFL), posterior facet length (PFL), and notch depth (ND) of the patients

Radiological Parameter	Mean Difference $\pm$ SD	95% CI of Mean Difference	Power (effect size d)
FNV	3.96 $\pm$ 1.24	(1.48) - (6.44)	0.99 (0.79)
APFA	5.67 $\pm$ 2.24	(1.18) - (10.16)	0.97 (0.64)
AFL	0.12 $\pm$ 0.63	(-0.38) - (1.71)	<80 (0.29)
PFL	0.66 $\pm$ 0.62	(-0.57) - (1.89)	<80 (0.27)
ND	-0.46 $\pm$ 0.28	(-1.03) - (0.1)	<80 (0.3)

In their study, Scranton et al (11) defined the axial malleolar index. They found that chronic ankle instability was associated with increased axial malleolar index angle. As the FNV angle increases in our study, the fibular notch turns more posteriorly than the tibial plafond. Therefore, the distal fibula is located more posteriorly in patients with high FNV than those with normal FNV values. In this respect, the 2 studies are similar. However, there is a problem in using the axial malleolar index to assess fibular position. Axial malleolar index measurement is related to the talus position and increases with internal talar rotation despite the same fibula position. In addition, Hu et al (22) in their biomechanical study, showed that when the ankle is brought from a neutral position or from 10° dorsiflexion to 15° plantar flexion under 600 N load, the distal fibula moves medially and anteriorly and rotates internally relative to the tibia. The same study showed that when the ankle position is changed from a neutral position or from 15° plantar flexion to 10° dorsiflexion, the fibula moves laterally and posteriorly and externally rotates relative to the tibia. The same study showed that the fibula made similar movements in different positions without axial loading on the ankle, but it was not the same as the amount of movement in axial loading. For this reason, we think studies based on the talus are inappropriate when investigating the relationship between ankle lateral ligament injury and distal tibiofibular joint. We believe that the axial malleolar index angles obtained from MR images taken under axial load will differ from the MRI measurement data taken without axial load. Our study has an advantage in terms of repeatability and consistency in that FNV measurement is not affected by any ankle or axial loading position.

LeBrun et al (23) defined the intermalleolar angle in their study. Since the internal rotation of the talus affects the measurement, the reference line based on the tibia was accepted as the medial malleolar articular surface. The distal fibula's position was determined relative to the anterior of the medial malleolus, but no significant relationship was found between the intermalleolar angle and ATFL rupture in the study. One of the reasons for this may be the decrease in angular sensitivity because the measurements made from the medial malleolus anterior to the fibula, which is taken as a reference, are made from a relatively long distance. The second reason may be that the medial malleolus is referenced by the line drawn from the articular surface, and variations of the medial malleolus affect the measurement. For a third reason, the posterior displacement of the anterior point of the medial malleolus can be considered as one moving distally from the tibiotalar joint level in the axial section. In our study, FNV measurement was made from 10 mm proximal to the tibiotalar joint and is a measurement made only within the tibia. Therefore, it is not possible to be affected by the distal fibula and medial malleolar positions. Therefore, it is a more consistent measurement.

The review article published by Noailles et al (24) examined the results of at least 10 y of patients who underwent anatomic and nonanatomical repairs due to lateral ankle instability. They reported better functional outcomes after anatomical repair, but recurrent instability was more frequent. In addition, they observed a range of motion (ROM)

loss and increased frequency of secondary osteoarthritis after nonanatomical repairs. The more frequent recurrent instability after anatomic repair may be due to differences in fibular notch anatomy. Although we do not have any data, differences in FNV may affect the results of primary repair. Further studies are needed to understand whether the fibular notch anatomy affects the success of conservative treatment in lateral ankle instability or the choice of method in patients scheduled for surgery.

A limitation to this study is the relatively low number of patients, retrospective nature, and the inclusion of patients with isolated ATFL rupture. We think 3-dimensional radiological imaging and biomechanical studies are needed to better understand the relationship between fibular notch anterior-posterior facet angle and ankle lateral complex injury.

In conclusion, a more posterior (retroverted) fibular notch and a lower fibular notch angle seem to be associated with higher rates of ATFL rupture.

Ethical Approval

Ethical approval was obtained from "Necmettin Erbakan University Ethical Committee"(2022/3572)

Informed Consent

Informed consent was obtained from all patients that their radiological images would be used for scientific purposes in accordance with the decision of the university ethics committee.

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